Option A: Vulnerability Deep-Dive (CVE of the Day)

CVE-2026-31431: “Copy Fail” Linux Kernel Local Privilege Escalation Vulnerability.

Date: 23 July 2026

Severity: High

Affected Technology: Linux Kernel

Vulnerability Type: Local Privilege Escalation

CVE: CVE-2026-31431

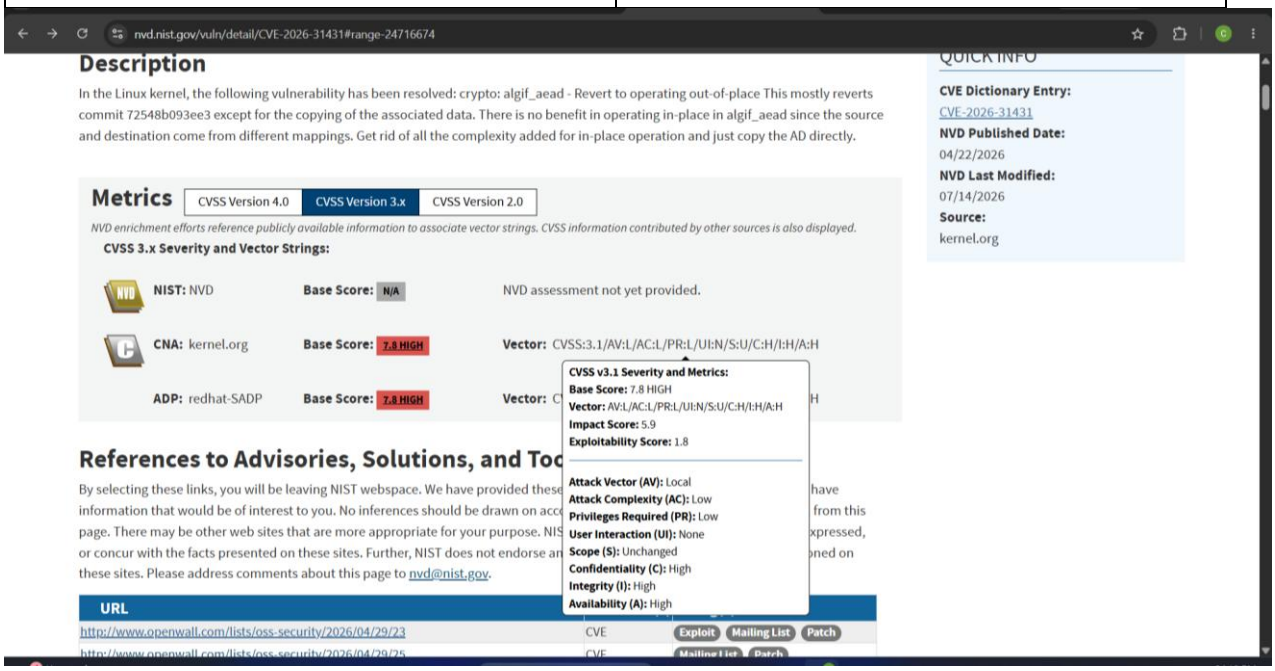
CVSS Score: 7.8 (High)

Research Questions:

1. Vulnerability Overview

Attribute	Details
CVE ID	CVE-2026-31431
Vulnerability Name	Copy fail
Affected product	Linux kernel
Vulnerability type	Local privilege escalation
CWE	CWE-669: Incorrect Resource Transfer between spheres
CVSS Score	7.8
Severity	High
Attack Complexity	Low
Privileges required	Low

User interaction	None
Confidentiality Impact	High
Integrity Impact	High
Availability Imapct	High



2. Root Cause Analysis

Root Cause Analysis

- The vulnerability is caused by improper handling of resources in the Linux Kernel's algif_aead component. The kernel does not correctly validate and manage some internal operations during cryptographic processing. Because of this mistake, a local attacker can misuse the flaw to gain higher privileges.

Technical Explanation

- **Vulnerability Type:** Local Privilege Escalation (LPE)
- **Root Cause:** Improper resource handling and insufficient validation of internal operations.
- **Related CWE:**
    - CWE-669 – Incorrect Resource Transfer Between Spheres

- CWE-1288 – Improper Validation of Consistency within Input

Weakness Enumeration

CWE-ID	CWE Name	Source
CWE-669	Incorrect Resource Transfer Between Spheres	CISA-ADP
CWE-1288	Improper Validation of Consistency within Input	redhat-SADP

Impact

If an attacker has local access to the system, they can exploit this vulnerability to gain **root privileges**. After getting root access, the attacker can control the system, access sensitive data, and make unauthorized changes.

3. Attack Vector

The vulnerability can be exploited **locally**. The attacker must have access to the target Linux system with a valid low-privileged user account. It cannot be exploited directly over the network. After accessing the system, the attacker can exploit the vulnerability to gain **root privileges**.

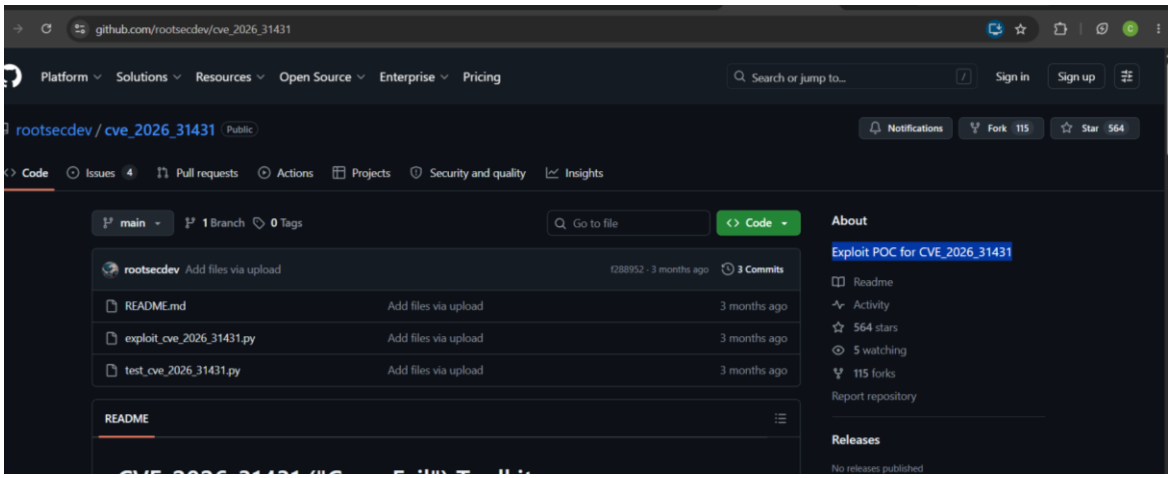
Attack Details

- **Attack Vector:** Local (AV:L)
- **Privileges Required:** Low (PR:L)
- **Network Exploitation:** No
- **User Interaction:** Not Required (UI:N)
- **Result:** Privilege Escalation to Root

4. Exploitation Status

- Public PoC: Publicly available Proof-of-Concept (PoC) exploit code is available for CVE-2026-31431. Multiple public repositories demonstrate exploitation of the vulnerability.
- Exploitation in the Wild: CVE-2026-31431 has been reported as actively exploited and is referenced in the CISA Known Exploited Vulnerabilities (KEV) Catalog.

- CISA KEV Status: Listed in the CISA Known Exploited Vulnerabilities (KEV) Catalog. Organizations should prioritize remediation according to CISA guidance.



5.Detection Opportunities

Security teams should monitor Linux systems for signs of local privilege escalation attempts related to CVE-2026-31431. The following activities may indicate suspicious behaviour:

- Unexpected processes gaining root privileges from a low-privileged user.
- Unusual activity involving the Linux AF_ALG cryptographic interface or the algif_aead component.
- Unexpected execution of privileged or setuid binaries.
- Suspicious modifications to privileged system files.
- Unusual process or authentication activity immediately before or after a user gains root privileges.

Organizations should correlate authentication logs, process execution logs, kernel-related events, and file-integrity monitoring to identify suspicious activity. Detection rules should be adapted to the specific Linux distribution and monitoring tools used by the organization.

6. Mitigation

- Patch Details:- Update the affected Linux Kernel to the latest security-fixed version provided by the relevant Linux distribution vendor. Reboot the system if required to load the patched kernel.

- Workaround:- If immediate patching is not possible, restrict untrusted local access to vulnerable systems and monitor for suspicious privilege-escalation activity until the security update can be applied.
- Vendor Guidance:- Organizations should follow the official security advisory and patch guidance provided by their Linux distribution vendor, such as Ubuntu, Red Hat, or other affected distributions. The exact fixed package and kernel version may vary depending on the Linux distribution and version in use.

7. References

- National Vulnerability Database (NVD) – CVE-2026-31431

https://in.search.yahoo.com/search?fr=mcafee&type=E210IN826G0&p=CVE-2026-31431+NVD

- Linux Kernel official security advisory and patch references

https://www.sophos.com/en-us/security-advisories/sophos-sa-20260501-copyfail

- Theori public Proof-of-Concept (PoC) repository

https://github.com/theori-io/copy-fail-CVE-2026-31431